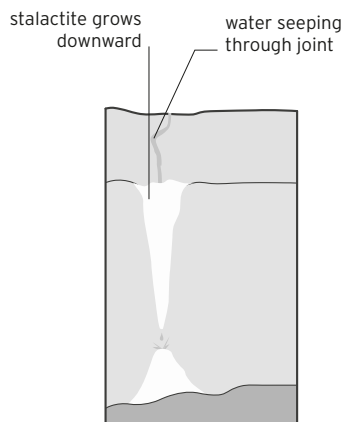
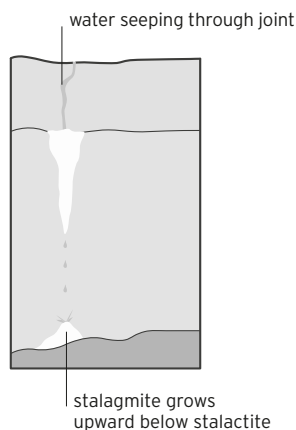


**STALACTITE ▼** A deposit, usually of calcite, hanging from the roof of a cave. Stalactites are produced when minerals precipitate from dripping water. See also *calcite*, *karst*, *precipitate*, *stalagmite*.



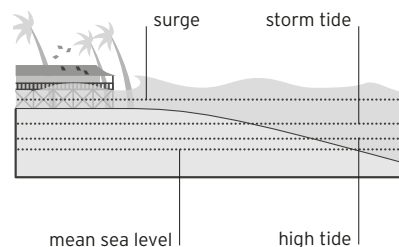
**STALAGMITE ▼** A deposit, usually of calcite, rising up from the floor of a cave. Like stalactites, stalagmites are produced when minerals precipitate from dripping water. See also *calcite*, *karst*, *precipitate*, *stalactite*.



**STAR DUNE** A pyramid-shaped mound of sand with three or more “arms” radiating from the center. Star dunes usually form where the wind blows from several directions, and they tend to grow up rather than spread out. See also *dune* (panel).

**STEPPE** A temperate grassland plain with very few trees, especially in regions with hot, dry summers and cold winters.

**STORM SURGE ▼** An abnormal rise of sea level, higher than the predicted tide, generated by a storm. The surge is caused by the storm’s winds pushing water onshore.

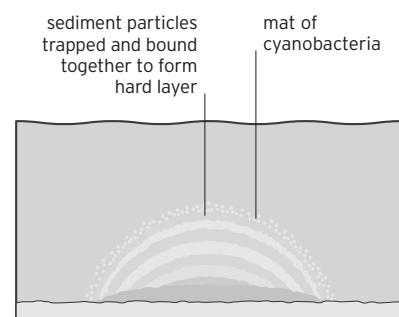


**STRATOSPHERE** The layer of the atmosphere between the top of the troposphere, at 5–10 miles (8–16 km), and the bottom of the mesosphere, at about 30 miles (50 km). See also *mesosphere*, *troposphere*.

**STRATOVOLCANO** A conical volcano composed of many layers of ash, pumice, and solidified lava. Unlike a shield volcano, its eruptions are usually violent. Stratovolcanoes are common near subduction zones. See also *pumice*, *subduction*, *volcano* (panel).

**STRATUM (PL. STRATA)** A layer of sedimentary rock. See also *bedding plane*, *sedimentary rock*.

**STRIKE-SLIP FAULT** A near-vertical fault with horizontal displacement of the rocks on either side. The San Andreas Fault in California is a strike-slip fault. See also *fault* (panel).



**STROMATOLITE ▲** A layered mound, column, or sheet formed by layers of single-celled cyanobacteria in ancient seas. When cyanobacteria

became common about 2.5 billion years ago, the oxygen they produced through photosynthesis began to change the atmosphere in a way that enabled other life forms to develop.

**STROMBOLIAN ERUPTION** A short-lived explosive volcanic eruption characterized by the ejection of ash, lava, and volcanic bombs. See also *phreatic eruption*.

**SUBDUCTION** The convergence of two tectonic plates, as a result of which an oceanic plate descends beneath another plate. Subduction zones can be classified as either ocean–ocean or ocean–continent, depending upon the nature of the two converging plates. See also *boundary* (panel), *deep-sea trench*, *tectonic plate*.

**SUBLIMATION** The change in a substance, such as glacier ice, from a solid to a gas without passing through the liquid phase.

**SUCCESSION** The gradual change from one community of plants to another over time in a particular area. For example, grassland may change to forest if grazing animals are removed from it.

**SUPERCCELL CLOUD** A large cloud characterized by the presence of a deep, persistently rotating updraft (a mesocyclone), lightning, torrential rain and hail, strong winds, and sometimes tornadoes. See also *thunderstorm*, *tornado*.

**SURGE-TYPE GLACIER** A glacier whose movement features periods of normal flow alternating with episodes of much more rapid flow, up to 100 times faster. The fast–slow cycle varies greatly between different surge-type glaciers. See *glacier*.

**SURTSEYAN ERUPTION** A volcanic eruption that takes place in relatively shallow water, with magma or lava interacting explosively with the

water. Surtseyan eruptions usually take place when an undersea volcano has finally grown large enough to break the water’s surface. See also *phreatic eruption*.

**SWAMP** A freshwater or saltwater wetland that is dominated by trees. See also *mangrove swamp*, *marsh*.

**SWASH** The rush of seawater up a beach when a wave breaks.

**SYMMETRICAL FOLD** A geological structure in which once-horizontal layers of rock have been folded in such a way that the fold’s axis is vertical. See also *asymmetrical fold*, *fold* (panel).

**SYNCLINE** A fold of stratified rocks in which the newest strata (layers) are at the core. Synclines result from massive horizontal compression. See also *anticline*, *fold* (panel).

## T

**TAIGA** The moist coniferous forest (boreal forest) biome covering much of northern Europe, Asia, and North America, south of the tundra. See also *biome*, *coniferous*, *tundra*.

**TECTONIC PLATE** One of the large, rigid sections into which Earth’s lithosphere is divided. Most earthquakes and volcanic activity are driven by the relative motion of plates. See also *lithosphere* (panel).

**TEKTITE** A glassy particle sometimes formed when a large meteorite strikes the Earth. Tektites are thought to have been created when a meteorite impact melted or vaporized surface rocks upon impact.

**TEMPERATE** Relating to those parts of Earth between the tropics and the polar regions. See also *tropical*.